

# Chankyu (Charlie) Han

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## RESEARCH INTERESTS

Smart Sensing and Haptic with AI/ML, Tactile Sensing, Human-Computer Interaction, Fabrication

## EDUCATION

### University of Washington (UW)

Seattle, Washington

Ph.D. in Electrical and Computer Engineering, Advisor: Dr. Yiyue Luo

Jun.2025 - Present

### Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Korea

M.S. in Mechanical Engineering, Advisor: Dr. Inkyu Park

Mar.2021 - Feb.2023

B.S. in Mechanical Engineering

Sep.2016 - Feb.2021

Graduation with *Cum Laude*

## PUBLICATIONS

(<sup>†</sup>: equal contribution)

### [11] [MagBall: Magnetic Rollerball for Multi-Scale Contact Interactions on Diverse Surfaces](#)

**C. Han**, Y. Miao, S. Zhang, Y. Luo

*CHI 2026: ACM Conference on Human Factors in Computing Systems* (accepted)

### [10] [Thermoforming 2D Films into 3D Electronics for Customizable Tactile Sensing](#)

J. Choi<sup>†</sup>, **C. Han**<sup>†</sup>, D. Lee<sup>†</sup>, H. Kim, G. Lee, J.-H. Ha, Y. Jeong, J. Ahn, H. Park, H. Han, S. Cho, J. Gu, I. Park

*Science Advances* 11.20 (2025): [eadv0057](#). <https://doi.org/10.1126/sciadv.adv0057>

### [9] [Spike-based Self-calibration for Enhanced Accuracy in Self-powered Pressure Sensing](#)

**C. Han**<sup>†</sup>, J. Choi<sup>†</sup>, J. Ahn, H. Kim, J.H. Ha, H. Han, S. Cho, Y. Jeong, J. Gu, I. Park

*Advanced Materials Technologies* (2023): [2301199](#). <https://doi.org/10.1002/admt.202301199>

### [8] [Recent Advances in Sensor-Actuator Hybrid Soft Systems: Core Advantages, Intelligent Applications, and Future Perspectives](#)

**C. Han**<sup>†</sup>, Y. Jeong<sup>†</sup>, J. Ahn<sup>†</sup>, T. Kim, J. Choi, J.-H. Ha, H. Kim, S.H. Hwang, S. Jeon, J. Ahn, J.T. Hong, J.J. Kim, J.H. Jeong, I. Park

*Advanced Science* (2023): [2302775](#). <https://doi.org/10.1002/advs.202302775>

### [7] [Nanotransfer Printing of Functional Nanomaterials on Electrospun Fibers for Wearable Healthcare Applications](#)

J.-H. Ha, J. Ko, J. Ahn, Y. Jeong, J. Ahn, S. Hwang, S. Jeon, D. Kim, S. A. Park, J. Gu, J. Choi, H. Han, **C. Han**, B. Kang, B.-H. Kang, S. Cho, Y.J. Kwon, C. Kim, S. Choi, G.-D. Sim, J.-H. Jeong, I. Park

*Advanced Functional Materials* (2024): [2401404](#). <https://doi.org/10.1002/adfm.202401404>

### [6] [Wireless, Multimodal Sensors for Continuous Measurement of Pressure, Temperature, and Hydration of Patients in Wheelchair](#)

S. Cho, H. Han, H. Park, S.-U. Lee, J.-H. Kim, S.W. Jeon, M. Wang, R. Avila, Z. Xie, K. Ko, M. Park, J. Lee, M. Choi, J.-S. Lee, W.G. Min, B.-J. Lee, S. Lee, J. Choi, J. Gu, J. Park, M.S. Kim, J. Ahn, O. Gul,

**C. Han**, K. Lee, S. Kim, K. Kim, J. Kim, C.-M. Kang, J. Koo, S.S. Kwak, S. Kim, D.Y. Choi, S. Jeon, H.J. Sung, Y.B. Park, Y.T. Choi, M. Je, Y.S. Oh, I. Park

*npj Flexible Electronics* 7.1 (2023): 8. <https://doi.org/10.1038/s41528-023-00238-3>

- [5] [Battery-Free, Wireless, Ionic Liquid Sensor Arrays to Monitor Pressure and Temperature of Patients in Bed and Wheelchair](#)

H. Han, Y.S. Oh, S. Cho, H. Park, S.-U. Lee, K. Ko, J.-M. Park, J. Choi, J.-H. Ha, **C. Han**, Z. Zhao, Z. Liu, Z. Xie, J.-S. Lee, W.G. Min, B.-J. Lee, J. Koo, D.Y. Choi, M. Je, J.-Y. Sun, I. Park

*Small* 19.9 (2023): 2205048. <https://doi.org/10.1002/smll.202205048>

- [4] [All-Recyclable Triboelectric Nanogenerator for Sustainable Ocean Monitoring Systems](#)

J. Ahn<sup>†</sup>, J.-S. Kim<sup>†</sup>, Y. Jeong, S. Hwang, H. Yoo, Y. Jeong, J. Gu, M. Mahato, J. Ko, S. Jeong, J.-H. Ha, H.-S. Seo, J. Choi, M. Kang, **C. Han**, Y. Cho, C. H. Lee, J.-H. Jeong, I.-K. Oh, I. Park

*Advanced Energy Materials* (2022): 2201341. <https://doi.org/10.1002/aenm.202201341>

- [3] [A Review of Recent Advances in Electrically-Driven Polymer-based Flexible Actuators: Smart Materials, Structures, and Their Application](#)

J. Ahn<sup>†</sup>, J. Gu<sup>†</sup>, J. Choi<sup>†</sup>, **C. Han**, Y. Jeong, J. Park, S. Cho, Y. Oh, J.-H. Jeong, M. Amjadi, I. Park

*Advanced Materials Technologies* (2022): 2200041. <https://doi.org/10.1002/admt.202200041>

- [2] [Stretchable Printed Circuit Board Based on Leak-Free Liquid Metal Interconnection and Local Strain Control](#)

M. S. Kim, S. Kim, J. Choi, S. Kim, **C. Han**, Y. Lee, Y. Jung, J. Park, S. Oh, B.-S. Bae, H. Lim, I. Park

*ACS Applied Materials & Interfaces* 14.1 (2021): 1826-1837. <https://doi.org/10.1021/acsami.1c16177>

- [1] [Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming](#)

J. Choi, **C. Han**, S. Cho, K. Kim, J. Ahn, D. D. Orbe, I. Cho, Z. Zhao, Y. Oh, H. Hong, S. S. Kim, I. Park

*Science Advances* 7.42 (2021): eabj0694. <https://doi.org/10.1126/sciadv.abj0694>

## TECHNOLOGY TRANSFER

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Joint transfer of the following technologies (\$37,000, 2022):

Finger wearable touch sensor, control system using same, and method thereof

I. Park<sup>†</sup>, J. Choi<sup>†</sup>, **C. Han**<sup>†</sup>, G. Lee<sup>†</sup>

*Korea Patent (Application Number: 10-2021-0096964), 2021.*

Large-Area Multi-Pressure Sensor Pad Fabrication Technology (Know-how)

I. Park<sup>†</sup>, J. Choi<sup>†</sup>, **C. Han**<sup>†</sup>, G. Lee<sup>†</sup>

## RESEARCH EXPERIENCES

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**UW Wearable Intelligence Lab**

Jun.2025 - Present

Research Assistant

Mentor: Yiyue Luo

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|----------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Korea University Sensor and Nanofabrication System Lab</b><br>Visiting Researcher                           | Jan.2025 – Jun.2025 |
| Mentors: Junseong Ahn, Junrak Choi                                                                             |                     |
| <b>KAIST Machine Intelligence and Novel-sensor Technology Lab</b><br>Undergraduate Intern & Research Assistant | Jun.2020 – Jun.2023 |
| Mentor: Inkyu Park                                                                                             |                     |

## SCHOLARSHIPS & AWARDS

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Best Oral Presentation Award</b><br>IEEE NEMS 2023 [9]                                                                                                                | May 2023               |
| <b>Best Poster Award</b><br>International Conference on Advanced Electromaterials (ICAE) 2021 [1]                                                                        | Nov. 2021              |
| <b>National Excellence Scholarship (Graduate Program)</b><br>Government funding for admission fees, full amount of school support fees in the master's program           | Mar.2021 – Feb.2023    |
| <b>KAIST Undergraduate Research Grant</b><br>Title: Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming [1]   | Sept. 2020 - Feb. 2021 |
| <b>Dean's List, School of Freshman</b><br>Top 2% of KAIST freshman cohort                                                                                                | Feb. 2017              |
| <b>National Excellence Scholarship (Undergraduate Program)</b><br>Government funding for admission fees, full amount of school support fees in the undergraduate program | Sep.2016 – Feb.2020    |

## SERVICE

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|----------------------------------------------------------------------------------------------------------------|----------|
| <b>Reviewer</b><br>Nature Communications (1), CHI 2026                                                         |          |
| <b>Outreach and Leadership</b><br>Workshop Organizer: Everyday Wearable for Personalized Health and Well-Being | CHI 2026 |
| High School Outreach: Talk and Lab Tour on Wearable Intelligence                                               | 2025     |

## SKILLS

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**Languages:** Python, C/C++, MATLAB, LabVIEW  
**Fabrication:** 3D printing (DLP, FDM), Thermoforming, Laser processing(CO<sub>2</sub>, UV, Fiber), PCB design  
**TOEFL:** 108/120

## OTHERS

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|--------------------------------------------|---------------------|
| Military Service in Republic of Korea Army | Jun.2023 – Dec.2024 |
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